ECO 362 Financial Investments
Precept 1: Discounting

September 8 – 12, 2025

Precept

- Review materials covered in lectures over the **previous week**.
- Q&A, Excel practice
- Precept slides in the same module as the lecture slides.
- You can attend any precept by any preceptor.
- We will maintain consistency across precepts (same slides).

Office Hours

- You can ask questions about anything related to the course.
- Prof. Yogo's OH is on Thursday.
- Preceptors' office hours are posted on Ed Discussion.
- Peer-tutors are available for individualized help.
- It is in your best interest not to leave the bulk of the work for the day before homework is due or for the week before the exam.

Online Discussion Board

- Besides precepts and OHs, you can post your question on Ed Discussions. You can post anonymously if you wish.
- We promise to check the discussion board each weekday to give timely answers.
- You are also encouraged to answer the question of your peers.
- Please use Ed Discussions instead of emails to avoid duplicates. Emailed questions will generally not be answered.

Outline

Q1: Start with \$1 today, how much wealth W_T do you have in T years?

- Suppose that the **annualized** interest rate is $Y = 12\%$.
  - **Convention:** time will be measured in years; $T = \frac{1}{2}$ means 6 months
- **Answer:** depends on the frequency of compounding (n)
  - semi-annually ($n = 2$) at 6% VS. monthly ($n = 12$) at 1%

$$W_T(n) = \left(1 + \frac{Y}{n}\right)^{nT}$$

Q: Is $W_T(2)$ larger or smaller than $W_T(12)$?

  - Continuous compounding ($n \to \infty$):

$$W_T(\infty) = e^{YT}$$

Discounting

Q2: To have \$1 in T years, how much wealth W_0 do you need to invest today?

- The answer depends again on the frequency of discounting:

$$W_0(n) \cdot (1 + Y/n)^{nT} = 1 \Rightarrow W_0(n) = \frac{1}{(1 + Y/n)^{nT}}$$

- Continuously compounded ($n \to \infty$):

$$W_0 \cdot e^{YT} = 1 \Rightarrow W_0 = e^{-YT}.$$

Doubling Time

Q3: How many years T_D until your 1 \$ doubles?

- Solve for T_D:

$$(1+Y/n)^{nT_D} = 2 \Rightarrow T_D(n) = \frac{1}{n}\frac{\ln(2)}{\ln(1+Y/n)}$$

- As $n \to \infty$:

$$e^{YT} = 2 \Rightarrow T_D = \frac{1}{Y}\ln(2) \approx \frac{72\%}{Y} \quad \textbf{"Rule of 72"}$$

Q: More frequent compounding means more or less time to double?

Outline

What is a Bond?

A bond is a particular *stream of cash flows* $\{X_t\}$ for $t = 1, \dots, T$.

- For now, assume these cash flows are *riskless*(no default).
- A bond *matures* at date T when it pays its final cash flow, called the *face/par* value

Bond examples: maturity of 3 years and face value $F = 100\$$

- **Zero-coupon bonds:**

$$X_1 = X_2 = 0, \quad X_3 = 100\$$$

- **Annuities:**

$$X_1 = X_2 = X_3 = 100\$$$

Yield of a Bond

Take a bond with cash flows $X_1, ..., X_T$ has price P today.

- Then the relation of the *annually compounded* **yield** Y and price P is:

$$P = \sum_{t=1}^{T} \frac{X_t}{(1+Y)^t}$$

- While for continuous compounding, obtain the yield y:

$$P = \sum_{t=1}^{T} X_t e^{-yt}$$

Definition:

- Given a frequency of compounding(annual/ continuous):
  the **yield** is the **discounting rate** such that the sum of discounted cash flows gives the price.
- Fixing the cash flows, the **yield** is simply a different way to quote the price, P; they are **inversely** related.

Zero-Coupon Bonds(ZCB): Yield

- Let $P_t^b(T)$ be the price of a ZCB of maturity T trading at time t, where we normalize the par to 1 \$ then:

$$P_t^b(T) = \frac{1}{(1 + Y_t^b(T))^T} \Longleftrightarrow Y_t^b(T) = \frac{1}{[P_t^b(T)]^{1/T}} - 1$$

$Y_t^b(T)$ denotes its yield *(annually compounded)*

- While the continuously compounded yield is given by:

$$P_t^b(T) \cdot e^{y_t(T) \cdot T} = 1 \iff y_t(T) = -\frac{ln(P_t(T))}{T}$$

ZCBs:Return

- A ZCB of maturity 3 trades in 2023 for $P^b_{2023}(3)$, while in 2024 it transforms into a ZCB of maturity 2 trading for $P^b_{2024}(2)$:

$$r^a_{2024} = \frac{P^b_{2024}(2)}{P^b_{2023}(3)} - 1$$

  is the annual return **realized** in 2024

- **Note:** $P^b_{2024}(2)$ is unknown at time $t = 2023$; this makes the annual return r^a_{2024} **risky**, even though the payment at maturity is *riskless*
- With continuous compounding:

$$e^{r_{2024}} \cdot P_{2023}(3) = P_{2024}(2) \iff r_{2024} = ln\left(P_{2024}\left(2\right)\right) - ln\left(P_{2023}\left(3\right)\right)$$

Yield Interpretation as Average of Returns

- Given a ZCB at t= 0, maturing at time T:
  - we know $y_0(T), P_0(T)$ and $P_T(0) = 1$;
  - $r_1^a, ..., r_T^a$ and $P_1(T-1), ...P_{T-1}(1)$ are not known (random) at period 0
- Under continuous compounding, the present yield equals the average of all future returns:

$$y_0(T) = \frac{1}{T}\sum_{t=1}^{T} r_t$$

"Discounting Examples" Example 2

Excel Demonstration

Outline

Long and Short Positions

- Recall that a bond is a stream of riskless cash flows: $X_1, ..., X_T$.
- Long(buy) the bond at price P in period 0 (at present).
    - Cash flows (including period 0): $-P, X_1, ..., X_T$.
- Short (sell) the bond at price P in period 0.
    - Cash flows (including period 0): $P, -X_1, ..., -X_T$.
- Short selling is equivalent to a loan.

Arbitrage

- Suppose Bond X with cash flows $X_1, ..., X_T$ traded at price P_X;
- Suppose Bond Y with cash flows $Y_1, ..., Y_T$ traded at price P_Y;

When does there exist an arbitrage opportunity?

- Two bonds (portfolios) with identical cash flows but different prices.
- Long cheap/low and short expensive/high.

Arbitrage: Example 1

- From now on, $X_t = 0$, $Y_t = 0$ if not mentioned.
- $P_X = 4, X_1 = 10;$
- $P_Y = 3$, $Y_1 = 10;$
- Yes.
  - Buy 1 unit of bond Y, short sell 1 unit of bond X.
  - Get \$1 today, no cash flow in the future.

Arbitrage: Example 2

- $P_X = 11, X_1 = 10, X_2 = 8;$
- $P_Y = 6$, $Y_1 = 5$, $Y_2 = 4$;
- Yes.
  - Buy 1 unit of bond X, short sell 2 units of bond Y.
  - Get $1 today, no cash flow in the future.

Arbitrage: Example 3

- $P_X = 0.1, X_1 = 10;$
- $P_Y = 5, Y_2 = 10;$
- No.

Outline

Law of One Price-No Arbitrage

Consider a bond with cash flows X_t, $t = 1, 2, \cdots, T$.

- The following portfolio gives you *identical* cash flows:
  - X_t units of t-year zero coupon bond, $t = 1, 2, \cdots, T$.
- The price of the bond consistent with no arbitrage is equal to the price:

$$\tilde{P} = \sum_{t=1}^{T} X_t P_b(t) = \sum_{t=1}^{T} \frac{X_t}{[1 + Y_b(t)]^t} = \sum_{t=1}^{T} X_t \cdot e^{-y(t) \cdot t}$$

However, the bond can trade at $P \neq \tilde{P} \Rightarrow$ Arbitrage Opportunity

- How would you take advantage of that opportunity (say if $\tilde{P} > P$)?

Main Idea:

- given the prices (or the yields) of ZCBs with different maturities T
- assuming no-arbitrage: $P = \tilde{P}$

, you can derive the price of any bond with arbitrary cash flows.

Application: Annuity

Consider T-year annuity with fixed payments 100\$ and price P.

Assuming non-arbitrage we apply the present value formula

$$P^a(T) = \sum_{t=1}^{T} \frac{100}{[1 + Y_t^b]^t}$$

Caution: do not confuse this with the price-yield relationship:

$$P^a(T) = \sum_{t=1}^{T} \frac{100}{(1 + Y^a(T))^t}$$

- $P_a(T)$ and $Y_a(T)$ are price and yield of this annuity (no time index).
- Y_t^b is the yield of the t-year zero-coupon bond (with time index).

"Discounting Examples" Example 3

Excel Demonstration